SOLUTION REQUIREMENTS

ComicCraft – AI Comic Story Creator using Gemini Models

Date	15 March 2026	**Team ID**	SWTID-2026-3296
Project Name	ComicCraft – AI Comic Story Creator using Gemini Models	**Team Size**	4

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The current ComicCraft version is designed for direct use without user account management. Future versions may add user login/profile support for personalized comic libraries.	Low
2	Authorization levels	The current version does not define separate user roles. A future admin/user model can be added for managing profiles, libraries, or administrative functions.	Low
3	External interfaces	The system integrates with Google Gemini models for structured comic outlines, narration and dialogue, and Hugging Face Diffusers / Stable Diffusion for comic-style image generation.	High
4	Transactions processing	The core transaction is the comic-generation workflow: receive user input, generate a panel outline, generate story narration/dialogue, generate panel images, build the layout, and export a PDF.	High
5	Reporting	The system should provide a comic preview and an exportable PDF containing panel titles, images, scene descriptions, captions and narration. The current project does not define business dashboards or analytics reports.	High
6	Business rules, etc.	User inputs include story prompt, main character name, setting, story tone and art style. The workflow generates a structured 5-panel outline, expands it into narration/dialogue, creates an illustration for each panel, and combines all outputs into a comic layout.	High
7	Compliance to laws or regulations	The project source does not specify a particular legal or industry compliance standard. API keys and tokens should be treated as sensitive configuration data and stored through environment variables rather than exposed in the interface.	Medium
8	Other	Provide a simple web interface, FastAPI routes for browser/API use, Jinja2 templates for dynamic rendering, a developer image-test route, and a PDF export success flow.	Medium

Step 2: Non-Functional Requirements (NFR)

Targets below are practical acceptance criteria for the documented local application workflow; the project source itself does not specify numeric SLA values.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The web interface should respond promptly to user actions and the backend should process the generation pipeline without unnecessary intermediate steps. Image generation and AI calls are the major processing stages.	Homepage and preview pages should load without blocking UI errors; generation should complete successfully for a valid request in the tested local environment.
2	Scalability	The architecture should allow independent scaling or replacement of the frontend, FastAPI backend, Gemini services and image-generation component. The current project is focused on local deployment.	The application should continue to support the documented generation flow as request volume grows, subject to available compute, API limits and model capacity.
3	Security & Data Privacy	Sensitive API credentials such as Gemini and Hugging Face keys should be stored in environment variables. User inputs and generated content should be handled through controlled backend routes.	No API keys hard-coded in templates or user-facing pages; required credentials are read from environment configuration.
4	Reliability & Availability	The application should gracefully handle API errors, JSON decoding errors and unexpected failures through try/except handling and FastAPI HTTPException responses.	Valid requests should complete without server crashes; failures should return a controlled error response or message.
5	Usability & Accessibility	The interface should provide clear labels, grouped input fields, readable panel-by-panel output, simple navigation, and a clear export-success flow. The current project emphasizes a clean and user-friendly web experience.	A user should be able to enter the required fields, generate the comic, review the preview, and reach the PDF export flow without needing technical knowledge.
6	Other – Maintainability & Portability	The modular project structure separates routes, Gemini integrations, image generation, layout building, export logic and templates, making components easier to test or replace.	Application should run in an isolated Python environment using the documented dependencies and local FastAPI/Uvicorn startup process.

Team Members
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